INCH-POUND

MIL-C-70570C (AR)
12 April 1993
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT BOARD, SERVO CONTROLLER: 9360875

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the fire control processor servo controller circuit card assembly referred to as the Servo Controller Board (SCB). The SCB is a 6 by 11 1/2 inch, double-layer, multiwire circuit board that is mounted vertically in the Fire Control Processor (FCP). The SCB inputs and outputs the necessary servo control parameters required by the M61A1 sight and gun/turret drives. Refer to Figure 1 for the SCB functional block diagram (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks for a part Of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and suPPlement thereto cited in the solicitation (see 6.2).

Beneficial comments **(recommendations, additions, deletions) and any** pertinent data which **may be** of **use in improving** this **document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document** or by letter.

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SPECIFICATIONS

MILITARY

MIL-F-13926 Fire Control Materiel Manufacture and Acquisition, Specification for

MIL-I-45607 Inspection, General Inspection and Disposition of Maintenance

STANDARDS

MILITARY

MIL-STD-lo9 Quality Assurance Terms and Definitions

MIL-STD-810 Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, handbooks are available from the Standardization Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2 2

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360875	-	Servo Controller Assembly
9360877	-	Servo Controller Schematic
9361337	-	Test Program Set
9397371	-	SIB/SCB, Interface Device TPS Adapter Board B, Printed Wiring Assembly
		Cable Assembly W1
9397497	-	Test Plug P6
9397523	-	Test Plug P7
9397524	-	Test Plug P8
9397525	-	Test Plug P9
9397526	-	Software Control Drawing
12563342	-	Requirements for Soldering Electrical and Electronic Assemblies
12950877	-	

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SPECIAL PACKAGING INSTRUCTION

PS 9360875 - Packaging of Servo Controller Assembly

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government Publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed **in the issue of the DODISS cited in** the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

ANSI/IPC-DW-425 - Design and End product Requirements for Discrete Wiring Boards

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018-3308.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. **Materials shall be in accordance** with the **applicable drawings and specifications.**

3.2 Components and assemblies. The components, and assemblies shall comply with all requirements specified on Drawing 9360875, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the SCB shall meet the requirements of MIL-F-13926 and ANSI/IPC-DW-425 as follows:

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3.2.1 MIL-F-13926.

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. part identification and marking.

d. Workmanship.

3.2.2 ANSI/IX-DW-425. IPC-DW-425/11 type Cl plated through connections.

3.3 Enviromental service conditions.

3.3.1 Vibration. The SCB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The SCB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three **shock** impulses of 15 g's 10%, 11 milliseconds + 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating. The SCB shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 TemPerature.

3.3.4.1 Operating. The SCB shall meet the requirements of 3.4 and 3.5 when stabilized the-lly at any temperature between **-400 and** +155°F.

3.3.4.2 Non-o~eratinu (storage). The SCB shall not suffer permanent damage when subjected to the constant temperature extremes of -600F for a period of 24 hours and +160"F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient .

3.3.5 Humidity (non-operatining) The SCB shall meet the requirements of 3.4 and 3. 5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days

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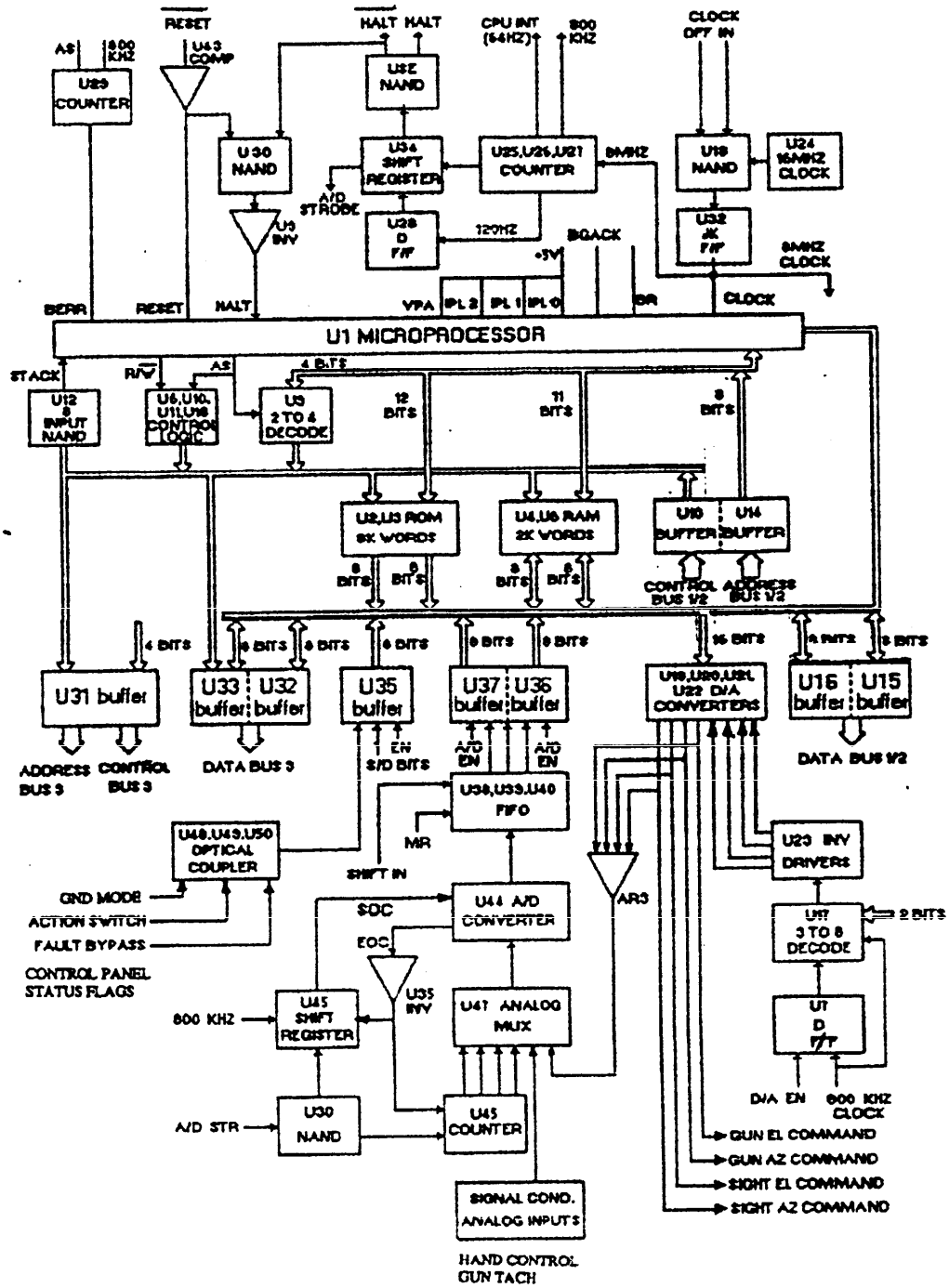


FIGURE 1. SCB functional block diagram.

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3.4 Electrical . Each conductor shall be electrically continuous as specified on the schematic diagram 9360877 of the SCB .

3.5 Performance.

3.5.1 Functional description. The SCB shall execute the Servo Control Processor (SCP) subprogram which controls all fire control, standby and BIT servo control modes. The SCB shall include control of input and output, (1/0) data timing, post-system command digital and analog filter processing, servo system communications with the Computer and memory board(CMB) Analog to Digital (A/D) and Digital to Analog (D/A) data conversion, seno system sampling rate timing, and control of the Synchro Interface Board (SIB). The SCB shall contain the primary 128 Hz and 64 Hz servo and communication clocks. The 128 Hz interrupt clock shall control SCP subprogram cycling, 1/0, and computation rates. The 64 Hz interrupt clock controls the SCB'S common memory. The SCB common memory shall consist of 4K bytes of RAM and 16K bytes of ROM. The SCB shall utilize analog to digital converters, digital to analog converters and an MC68000 microprocessor to manage the fire control and servo functions. Synchro-to-digital (S/D) position data, control panel status, and boresight angle data shall be input along with A/D data to the SCB . Output data in analog form shall pass through four D/A converters to the system gun and sight activators. Digital data shall output through buffers to the SIB for the digital-to-synchro (D/S) converters and to the CMB.

3.5.2 Power requirements. SCB power requirements shall be as listed in Table I.

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TABLE I. Power requirements.

	VOLTAGE (Vdc) MIN	TYP	MAX	AMPS Adc	IMPD OHMS	RIPPLE V p-p	INPUT CONN. PINS	RETURN CONN. PINS
+ 5 V	4.75	5.00	5.25	1.5	<.033	<.100	P1-32, 33, 34, 72, 73 and 74 P2-33 and 68	P1-35, 36, 37, 75, 76 and 77
+15 V	14.25	15.00	15.75	.533	<.033	<.100	P1-40 and 80	P1-38 and 78
-15 V	-14.25	-15.00	-15.75	.533	<.033	<.100	P1-39 and 79	P1-38 and 78

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3.5.3 TTL voltage levels. The TTL voltage levels for inputs and outputs are as follows:

	MIN	TYP	MAX
Voltage Output Low (VOL)	---	0.2	0.4
Voltage Output High (VOH)	**2.4**	**3.4**	**---**
Voltage Input Low (VIL)	---	----	0.7
Voltage Input High (VIH)	2.0	---	---

3.5.4 Oscillator and counting circuits.

3.5.4.1 Clock frequencies and tolerances. With all power supply voltages applied to the SCB per 3.5.2, the clock rates for all control logic and the MC68000 shall be generated by a counter circuit excited by 16 MHZ oscillator u24. The 16 MHz signal shall be divided by flip-flops (U52, U28) and dividing counters (u25, U26, U27) to produce the TTL signals and frequencies listed in Table II.

TABLE II. Frequency tolerance.

	MAX	MIN	UNITS	cam. Pm	sIamL NAME
16	16.0016	15.9984	MHz	JI-16, 17	16 MHz
8	8.0008	7.99920	MHz	J1-15	8 ~z
800	800.08	799.920	KHz	J1-14, PI-28	800 KHZ
3.2	3.20032	3.19968	KHz	J1-13	3.2 KHz
128	128.013	127.987	Hz	J1-8	128 Hz
64	64.0064	63.9936	Hz	Pi-63	CPU INT

3.5.4.2 External clock. A TTL loW INTERNAL CLOCK DISABLE signal (J1-1) and NAND gates (u18) shall prevent the oscillator from driving the SC9 during testing to allow the injection of an external clock frequency (EXT CLOCK J1-2) .

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3.5.4.3 Counting circuit derived signals.

3.5.4.3.1 A/D STR (J2-45) and~ (J2-23). The AID STR signal shall be a string of 250 ~sec pulses with a frequency of 128 Hz. The signal is generated & "D" flip-flop U28 driven by the 128 Hz clock. The U28 SET line (J2-9) is tied to TTL high by pull-up resistor U59 and is used to prevent U28 from being reset and to cause the TTL low-high transition of U28. The QA ou~ut of shift register U34, driven by the 3.2 KHz clock and the Q output of U28, shall reset U28 to shorten the pulsewidth to 250 * 0.1 psec. The A/D STR signal shall be generated by inverting AtD STR with U30. The A/D STR shall provide the functions as specified in the following paragraphs.

3.5.4.3.1.1 Reset FIFO resisters U38-U40. The high-to-low transition of the A/D ST'R (J2-23) signal shall reset First-In-First-Out (FIFO) registers U38-U40 to prepare them to receive data from A/D converter U44.

3.5.4.3.1.2 Reset counter U45. The high-to-low transition of A/D (J2-23) shall reset counter U45, setting multiplexer u47 address lines (QA, QB, QC, QD) TTL low.

3.5.4.3.1.3 Generate SOC (J2-21) signal. Six and one half 800 KHz clock cycles after receiving the low-to-high transition of the A/D STR (J2-23) signal, shift register U46, inverter U35 and flip-flop U28 shall force SOC (J2-21) TTL high for one 800 KHZ clock cycle, at which time U46 CLR (J2-30) IS forced low, resetting **U46.**

3.5.4.3.1.4 Generate CPU INT (P1-63). The CPU INT (E1-63) signal is a 64 Hz TTL square-wave signal generated by the Q output of U28 and counter U27 notifying the Computer and Memory Board that it may have access to **SCB RAM.**

3.5.4.3.1.5 Generate HALT RESET (J2-4). HALT RESET shall be generated by clocking Q output of **U28** through shift register U34. HALT RESET is a **312.5** psec pulse at a frequency of **128** Hz. The HALT RESET signal shall trigger **'S-R'** flip-flop U30 to generate the HALT (J2-6) signal.

3.5.5 Reset circuit. When power is applied to the SCB, the power-on reset **(U43, U62, R53, R54, R55, R62, D8 and C42) circuit shall** apply a TTL low to the MC68000 (Ul) RESET ($^{U1-18}$) **and HALT** (U1-17) pins (RESET J2-2 and RP HALT J2-1, respectlvely) . . . - 200 msec. With power applied to the MC68000 and a clock frequency of at least 4 MHz asserted on the MC68000 clock pin, he MC68000 shall tri-state its busses and place ~ts output

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control lines in their inactive state. When C42 charges to 2.5 ± .25 **Vdc,** the output of comparator U43 shall go TTL high, causing the MC68000 HALT and RESET pins to go inactive. This, in turn, causes the busses and control lines to go active. The MC68000 then executes a reset vector hardwired internally to the MC68000. This reset vector prefetches the program counter and stack pointer registers values from ROM, which begins execution of Initial Servos, a routine from the SCB ROM. A reset applied to J1-6 (RESET) shall discharge C42, initiating the power-on sequence.

3.5.6 I/O Address Decoder. The peripheral devices on the SCB shall be enabled from signals generated by the MC68000 and the peripheral control decoder. The AS (J1-11) signal shall be generated by the MC68000 and shall be inverted by U6 to generate the AS (J2-26) signal and inverted again by U6 to generate DEC EN (J2-27). $\overline{\text{DEC EN}}$ shall enable address decoder U9 which decodes address lines AS, A6, A14 and A15 (J1-30, 29, 21 and 20 respectively) . The outputs of the address decoder shall be as specified in Table III.

TABLE III. I/O addresses.

A15	A14	A6	As	SIGNAL	TEST POINT
0	0	X	X	$\overline{\text{ROM EN}}$	J1-19
0	1	X	X	$\overline{\text{RAMEN}}$	J1-18
1	0	0	0	$\overline{\text{DIA EN}}$	J2-48
1	0	0	1	$\overline{\text{SYNCHRO EN}}$	J2-47
1	0	1	0	$\overline{\text{AID EN}}$	J2-46
1	0	1	1	$\overline{\text{D/W EN}}$	J2-25
1	1	X	X	$\overline{\text{HALT EN}}$	J2-49

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3.5.6.1 ROM EN. When pulled to a TTL low state the ROM EN (J1-19) signal enable ROM chips U2 and U3. ROM EN shall also be inverted by U6 to generate ROM EN (J2-28), which is used to clear up the two UIO 'D" flip-flops. U1O SET (J2-5) is tied to +5 V by pull-up resister U62 to prevent U1O from being preset. The U1O flip-flops shall generate ROM DTACK (J2-13) between .250 jl.sec and .375 Usec after receiving ROM EN. ROM DTACK shall generate the DTACK (J2-12) signal through gate U12.

3.5.6.2 RAM EN. When pulled to a TTL low state, the RAM EN (J1-18) signal shall generate the DTACK signal through U12 and be used as inputs to two of U8'S tri-state gates. RAM EN shall also be inverted by U6 to generate RAM EN (J2-29). The RAM EN signal shall be used to prevent the two Ull "D" flip-flops from being preset. U1l-RESET (J1-12) shall prevent Ull from being cleared and is tied to +5 Vdc by pull-up resistor u62.

3.5.6.3 The - (J2-48) signal shall be used in conjunction with the 800 KHz CLK signal to enable decoder U17, which is used to select the D/A converter. D/A EN shall also be inverted by U23 to generate D/A EN (J2-19), which shall reset the two U7 'D" flip-flops. The U7 flip-flop preset pins shall be tied to a 'TTL high by U7 SET (J2-20) to prevent erroneous D/A DTACK (J2-35) signals" Within 1.25 Usec of receiving the D/A EN (J2-19) signal, the first flip-flop will generate Q (J2-18), which shall enable decoder U17. After one 800 KHz clock cycle, Q shall toggle low, disablina u17, and the second U7 flip-flop shall generate D/A DTACK. D/A DTACK shall generate the DTACK signal through gate u12.

3.5.6.4 SYNCHRO EN. The SYNCHRO EN (J2-47) signal shall be loaded into buffer U31 and passed out as SYN EN on Pi-29.

3.5.6.5 A/D EN. The (J2-46) signal shall be used to generate the DTACK signal through gate U12 and enable buffers u36 and u37, and to trigger the shift-out function of FIFO registers u38-U40.

3.5.6.6 DW EN. The DW EN (J2-25) signal shall be used to generate the DTACK Signal through gate U12 and to enable data buffer U55.

3.5.6.7 HALT EN. The HALT EN (J2-49) signal shall be used to generate the DTACK signal through gate U12 and to generate the HALT (J2-6) signal through "S-R' flip flop U30.

3.5.6.8 DTACK and BERR. The DTACK (J2-12) signal, shall be generated by the outputs of address decoder U9, SYNCHRO DTACK

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NANDed by gate U18 to produce the DTACK (J1-lO) signal which shall cause the MC68000 to toggle AS (J1-11) high. If DTACK is not **received by the MC68000 within 10** Usec of **the** generation of **X (JI-11), count=** u29 shall PrOdUCe BERR (J2-3), signaling the MC68000 of a bus error.

3.5.7 AID converter interface. The AID converter interface shall consist of S differential amplifiers (ARl, AR2), a summing amplifier (AR3), 4 low-pass filter channel a sixteen channl multiplexer (U47), a multiplexer address counter (U45), a 12 bit A/D converter (U44), 3 First-In-First-Out (FIFO) **data registers (u38, U39, U40), and associated control logic (U28, U35, U46) .**

3.5.7.1 Multiplexer U47 channels. Per Table IV, simultaneously applying each INPUT VOLTAGE to its respective nominal analog input pins and address word for the SCB U47 multiplexer shall cause the following outputs with values per Table IV.

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TABLE IV. Multiplexer U47 channels, inputs, outputs.

MUX CHANNEL	QD QC QB QA (J2-39,40,41,42)	INPUT SIGNAL	P2-PIN	INPUT VOLTAGE (Vdc)	MULTIPLEXER INPUT (Vdc)	MULTIPLEXER INPUT SIGNAL	P2-PIN	MUX OUTPUT (J2-43) A/D IN (Vdc)
S1	0 0 0 0	GROUND	--	-0.010 TO 0.010	-0.010 TO 0.010	--	--	-0.010 TO 0.010
S2	0 0 0 1	HAND CONTROL AZ	38 39	9.990 TO 10.010 -0.010 TO 0.010	-9.609 TO -9.701	MUX INPUT 1	57	-9.609 TO -9.701
	0 0 0 1	HAND CONTROL AZ	38 39	-0.010 TO 0.010 -9.990 TO -10.010	9.609 TO 9.701			9.609 TO 9.701
S3	0 0 1 0	HAND CONTROL AZ	38 39	1.490 TO 1.510 -0.010 TO 0.010	-9.609 TO -9.701	MUX INPUT 2	58	-9.609 TO -9.701
	0 0 1 0	HAND CONTROL AZ	38 39	-0.010 TO 0.010 -1.490 TO -1.510	9.609 TO 9.701			9.609 TO 9.701
S4	0 0 1 1	GUN TACH AZ	41 43 42	9.990 TO 10.010 -0.010 TO 0.010	-3.192 TO -3.329	MUX INPUT 3	59	-3.192 TO -3.329
S5	0 1 0 0	GUN TACH AZ	41 43 42	2.990 TO 3.010 -0.010 TO 0.010	6.808 TO 7.101	MUX INPUT 4	60	6.808 TO 7.101
S6	0 1 0 1	HAND CONTROL EL	3 4	9.990 TO 10.010 -0.010 TO 0.010	-9.609 TO -9.701	MUX INPUT 5	61	-9.609 TO -9.701
	0 1 0 1	HAND CONTROL EL	3 4	-0.010 TO 0.010 -9.990 TO -10.010	9.609 TO 9.701			9.609 TO 9.701
S7	0 1 1 0	HAND CONTROL EL	3 4	1.490 TO 1.510 -0.010 TO 0.010	-9.609 TO -9.701	MUX INPUT 6	62	-9.609 TO -9.701
	0 1 1 0	HAND CONTROL EL	3 4	-0.010 TO 0.010 -1.490 TO -1.510	9.609 TO 9.701			9.609 TO 9.701
S8	0 1 1 1	GUN TACH EL	6,8 7	9.990 TO 10.010 -0.010 TO 0.010	-1.517 TO -1.583	MUX INPUT 7	63	-1.517 TO -1.583

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TABLE IV. Multiplexer U47 channels, inputs, outputs. (continued)

MUX CHANNEL	QD QC QB QA (J2-39 - 42)	INPUT SIGNAL	P2-PIN	INPUT VOLTAGE (Vdc)	MULTIPLEXER INPUT (Vdc)	MULTIPLEXER INPUT SIGNAL	P2-PIN	MUX OUTPUT (J2-43) A/D IN (Vdc)
S9	1 0 0 0	GUN TACH EL	6, 8 7	2.990 TO 3.010 -0.010 TO 0.010	3.166 TO 3.452	MUX INPUT 8	64	3.166 TO 3.452
S10	1 0 0 1	GROUND	–	-0.010 TO 0.010	-0.010 TO 0.010	MUX INPUT 9	65	-0.010 TO 0.010
S11	1 0 1 0	GROUND	–	-0.010 TO 0.010	-0.010 TO 0.010	MUX INPUT 10	22	-0.010 TO 0.010
S12	1 0 1 1	MANUAL/EXT RANGE	12	14.990 TO 15.010	9.609 TO 9.701	MUX INPUT 11	23	9.609 TO 9.701
S13	1 1 0 0	CURRENT AZ 1	44	9.990 TO 10.010	9.609 TO 9.701	MUX INPUT 12	24	9.609 TO 9.701
S14	1 1 0 1	CURRENT AZ 2	46	9.990 TO 10.010	9.609 TO 9.701	MUX INPUT 13	25	9.609 TO 9.701
S15	1 1 1 0	CURRENT EL	9	9.990 TO 10.010	9.609 TO 9.701	MUX INPUT 14	26	9.609 TO 9.701
S16	1 1 1 1	DAC4	–	-4.969 TO -5.031 -0.010 TO 0.010 4.969 TO 5.031	4.839 TO 5.162 0.010 TO -0.010 -4.839 TO -5.162	MUX INPUT 15	27	4.834 TO 5.167 0.015 TO -0.015 -4.834 TO -5.167
		DAC3	–	-4.969 TO -5.031 -0.010 TO 0.010 4.969 TO 5.031	4.839 TO 5.162 0.010 TO -0.010 -4.839 TO -5.162			4.834 TO 5.167 0.015 TO -0.015 -4.834 TO -5.167
		DAC2	–	-4.959 TO -5.041 -0.010 TO 0.010 4.969 TO 5.041	4.829 TO 5.171 0.010 TO -0.010 -4.829 TO -5.171			4.824 TO 5.177 0.015 TO -0.015 -4.824 TO -5.177
		DAC1	–	-4.959 TO -5.041 -0.010 TO 0.010 4.959 TO 5.041	4.829 TO 5.171 0.010 TO -0.010 -4.829 TO -5.171			4.824 TO 5.177 0.015 TO -0.015 -4.824 TO -5.177

NOTES: 1. THE A/D DIGITAL OUTPUTS OF TABLE IV SHALL BE INVERTED BY BUFFERS U36 AND U37 BEFORE BEING APPLIED TO THE DATA BUS.
2. FOR S16 THE DAC'S ARE PROGRAMMED FOR -5.000, 0.000 AND +5.000 VDC NOMINALLY.

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TABLE IV. Multiplexer U47 channels, inputs, outputs. (continued)

MUX CHANNEL	ACCEPTABLE A/D DIGITAL OUTPUT RANGE FROM	TO
S1	000000001011	111111110011
S2	011110011000 100001100110	011111011000 100000100110
S3	011110011000 100001100110	011111011000 100000100110
S4	001001110110	001010111111
S5	101010100011	101000111011
S6	011110011000 100001100110	011111011000 100000100110
S7	011110011000 100001100110	011111011000 100000100110
S8	000100011111	000101011001
S9	110110001111	110100100110
S10	000000001011	111111110011
S11	000000001011	111111110011
S12	100001100110	100000100110
S13	100001100110	100000100110
S14	100001100110	100000100110
S15	100001100110	100000100110
S16	110000110111 111111110010 001111000111 110000110111 111111110010 001111000111 110000111001 111111110010	101111000111 000000001100 010000110111 101111000111 000000001100 010000110111 101111000101 000000001100

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3.5.7.2 A/D Converter. Timing shall be as in Figure 2. Upon receiving the trailing edge of a 1.25 - 1.00 psec SOC pulse, A/D converter U44 shall digitize the analog output of U47, A/D IN, into a 12 bit negative true logic complementary-two's-complement (CTC) coded data word (A108 **mV) as specified in Table V. Buffers** u36 **and U37 shall** invert the binary output shown in Table V before applying it to the **data** bus. During conversion **(10 usec mxJ** U44 shall output a TTL high (normally) TTL low) on pin U44-20. This signal shall be inverted by U35 to produce the signal E(X J2-22), which shall act as the clock input, incrementing counter U45. At the **end of** the conversion, U44-U20 shall return to a TTL low, toggling EOC to a TTL high, which shall trigger FIFO registers u38-u40 to shift in the digital word.

TABLE V. **AID transfer Characteristic**

Analog Input	binary utput MSB LSB	analog input	Binary output MSB LSB
10.000 v	100000000000	**-4.883 EN**	000000000000
5.000 v	101111111111	**-9.766 W**	000000000001
2.500 v	110111111111	-19.531 mv	000000000011
1.250 V	1111011111111	**-39.063** m V	000000000111
62S.000 mV	111101111111	-78.125 mV	000000001111
312.500 mV	111110111111	-156.250 mV	000000011111
156.250 mV	111111011111	-312.500 mV	000000111111
78.125 W	**111111101111**	**-625.000 mV**	000001111111
39.063 mV	111111110111	**-1.250 V**	000011111111
19.531 mV	111111111011	-2.500 v	000111111111
9.766 mV	111111111101	**-5.000 v**	001111111111
4.883 mv	111111111110	-10.000 v	011111111111
0.000 mV			

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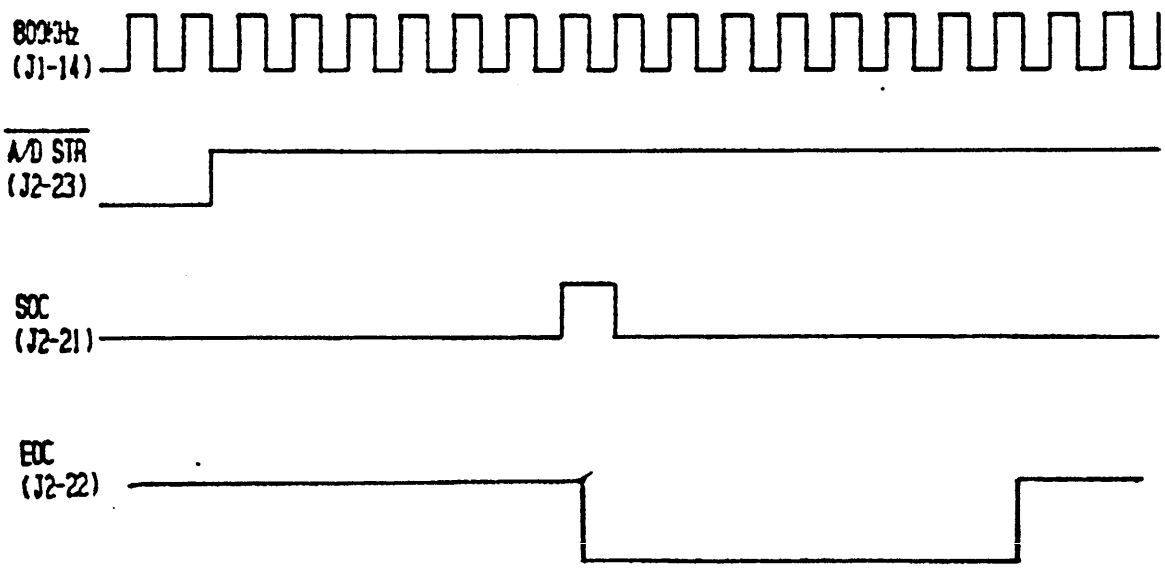


FIGURE 2. A/D conversion.

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3.5.8 Control panel status. Upon receipt of the DW EN {J2-25) signal, Control panel status shall be loaded into tri-state data buffer U55. SP/TWD, S/D l-BIT, S/D 2-BIT, S/D 3-BIT, S/D 4-BIT (P2-52, 53, 54, 55, 56, respectively) TTL inputs shall be maintained by the use of pull-up resistors U53 and U51. The GRouND MODE (P2-49), ACTiON SWITCH (P2-50) and FAULT BYPASS (P2-51) signals are O or +24 Vdc signals which shall be buffered by opto-isolators U48-U50 to produce the TTL signals GND MODE (J2-38), ACTION SWITCH (J2-36) and FAULT BYPASS (J2-37), respectively. U48-U50 shall accomplish this by producing a TTL high for inputs less than 0.8 Vdc and a TTL low for inputs greater than 1.2 Vdc.

3.5.9 **D/A converters. Upon receiving D/A EN (J2-48) low, Q (J2-18)** high, R/W (J1-9) low, and 800 KHz **CLK (J1-14) low,** address decoder u17 shall generate a data latch signal through one of U23'S inverters determined by address lines Al (J1-34) and A2 (J1-33) as specified in Table VI. The data latch signal shall latch the specified D/A to the 16-bit offset binary data word on data bus D15-DO (J1-35-50). Each D/A shall be capable of converting the specified data word into a bipolar analog output as specified in Table VII with tolerances as follows:

DAC		programmed to zero	

The output of each D/A converter shall also be used as an input for multiplexer U47 (through U58 and AR3) for End-Around testing.

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TABLE VI. D/A address, latch signals and output signals.

A2	A1	D/A	Latch Signal	Conn. Pin No.	Output Signal	Conn. Pin No.
0	0	U20	D/A 2LD	J2-32	SIGHT COMMAND EL HI SIGHT COMMAND EL LO	P2-34 P2-35
0	1	U22	D/A 4LD	J2-34	SIGHT COMMAND AZ HI SIGHT COMMAND AZ LO	P2-69 P2-70
1	0	U21	D/A 3LD	J2-33	GUN COMMAND EL HI GUN COMMAND EL LO	P2-31 P2-32
1	1	U19	D/A 1LD	J2-31	GUN COMMAND AZ HI GUN COMMAND AZ LO	P2-66 P2-67

TABLE VII. D/A transfer characteristics.

Binary Input D15-D0	Analog Output	Binary Input D15-D0	Analog Output
1111111111111111	10.000 V	0111111111111111	-305.176 µV
1100000000000000	5.000 V	0111111111111110	-610.352 µV
1010000000000000	2.500 V	0111111111111100	- 1.221 mV
1001000000000000	1.250 V	0111111111111000	- 2.441 mV
1000100000000000	625.000 mV	0111111111110000	- 4.883 mV
1000010000000000	312.500 mV	0111111111100000	- 9.766 mV
1000001000000000	156.250 mV	0111111111000000	- 19.531 mV
1000000100000000	78.125 mV	0111111110000000	- 39.063 mV
1000000010000000	39.063 mV	0111111100000000	- 78.125 mV
1000000001000000	19.531 mV	0111111000000000	-156.250 mV
1000000000100000	9.766 mV	0111110000000000	-312.500 mV
1000000000010000	4.883 mV	0111100000000000	-625.000 mV
1000000000001000	2.441 mV	0111000000000000	- 1.250 V
1000000000000100	1.221 mV	0110000000000000	- 2.500 V
1000000000000010	610.352 µV	0100000000000000	- 5.000 V
1000000000000001	305.176 µV	0000000000000000	- 10.000 V
1000000000000000	0.000 V		

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3.5.10 Halt control circuit. 'S-R' flip flop U30 shall generate the HALT (J2-6) and HALT signals, when receiving the HALT EN (J2-49) signal specified in Section 3.5.6.7. The HALT signal shall disable the RAM WRITE (J2-10) and RAM READ (J2-16) signals from being controlled by the MC68000 by using tri-state gates U8. HALT signal 1 enables data buffers U13 and U14, allowing RAM READ and RAM WRITE to be controlled by READ (PI-24) and 2 DELAY WRITE) , respectively " SCB P1 22 shall control the B1/2 EN (J2-11), RAM EN U (J2-15), 17) lines to enable 1/0 buffers u15 and u16, RAM U4, and RAM US, respectively. Through gates U30 and u35 HALT shall generate the signals PR HALT (J2-24) and PR HALT (J2-1), halting MC68000 program execution.

3.5.11 Memory requirments.

3.5.11.1 ROM. The SCB shall contain two ROM chips (U2, U3). Each chip shall be preprogrammed as specified on government drawing 12563342. U2 and U3 shall be capable of on-board reprogramming upon assertion of a TTL low on U2 PGM (P2-13) and U3 PGM (P2-14), respectively. U2 PGM and U3 PGM shall be tied to +5 Vdc by pull-up resistors on u51 to prevent accidental reprogramming. Timing and signal requirements for ROM read cycles shall be as specified in Figure 3.

3.5.11.2 FWYI. The SCB shall contain two RAM chips (U4, U5). Timing and signal requirements for RAM read and write cycles shall be as specified in Figure 4 and 5, respectively.

3.5.11.3 Common memory. The SCB shall allow an external Computer and Memory Board (CMB) t share SCB RAM, during the common memory window access time. The common memory wxndow shall open (begin) when CPU INT (PI-63) goes logic low and close (end) when PR HALT (J2-1) goes logic high. The access time window (interval) shall be 1.1875 ± 0.0003 msec long. The access time interval shall repeat at a 64 Hz rate.

3.5.12 Bus arbitration. SCB bus arbitration shall be accomlished by the BUS REQUEST (J1-s), BUS GRANT ACK (J1-4) and BUS GRANT (J1-S) signals. An external device desiring to control the bus shall assert a TTL low on the BUS REQUEST line. The MC68000 shall respond to the BUS REQUEST signal by generating a TTL low on BUS GRANT at the end of the present bus cycle. The

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requesting device shall respond by asserting a logic low on the BUS GRANT ACK line, negative (i.e., remove by making logic high) its BUS REQUEST signal; and take over as the new bus master. The MC68000 shall continue its processing **if** the BUS REQUEST is canceled before BUS GRANT ACk is received. To relinquish bus control back to the MC68000, the requesting device shall negate its BUS GRANT ACK **signal.**

3.6 Input~outPut requirments.

3.6.1 power supply **and around pins.** The SCB shall have **power supply and ground pins as specified** in table VIII.

3.6.2 Connector PI inputs. The SCB connector PI inputs shall be TTL per 3.5.3 with input pins as specified in Table IX.

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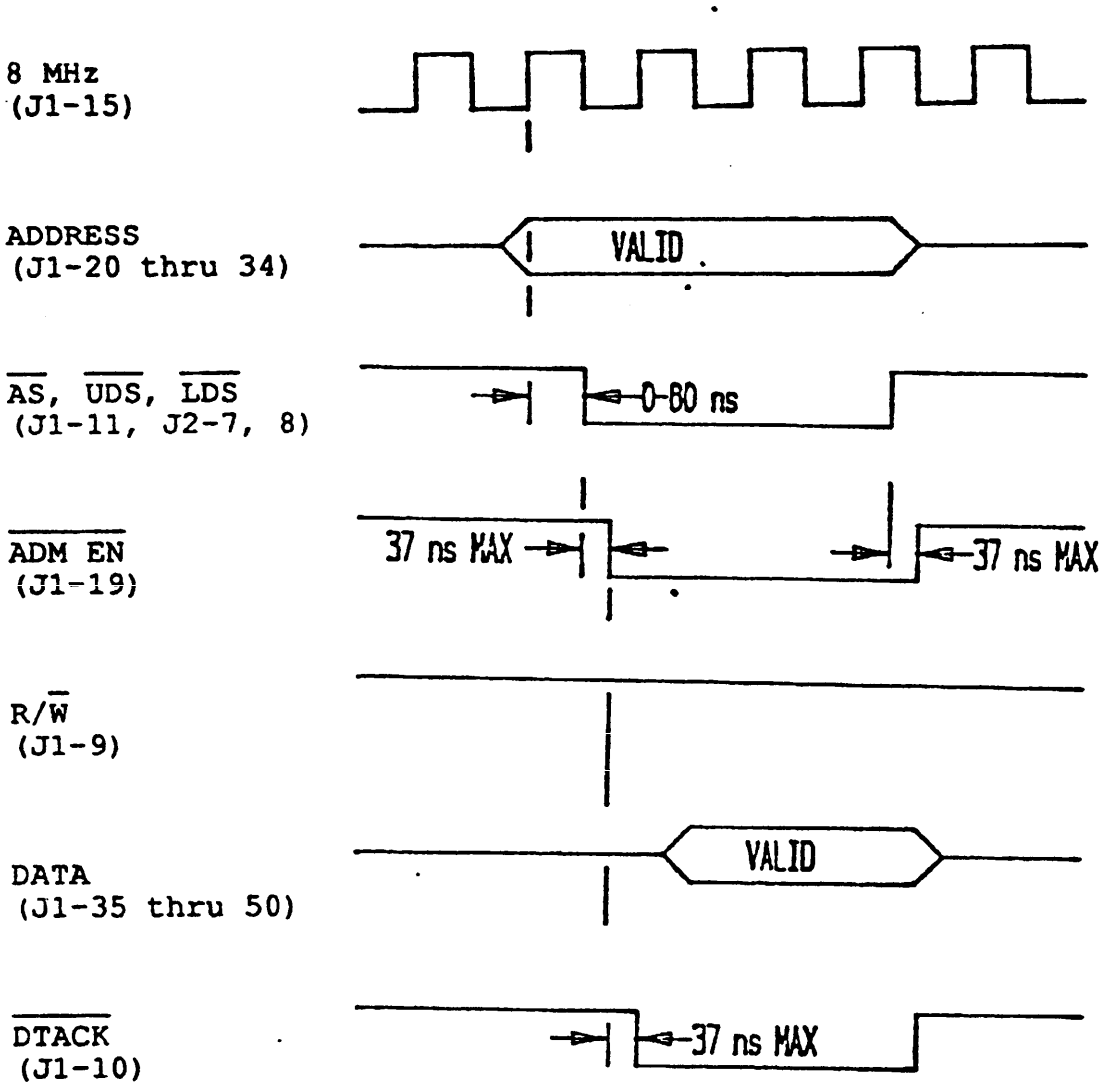


FIGURE 3. ROM read cycle.

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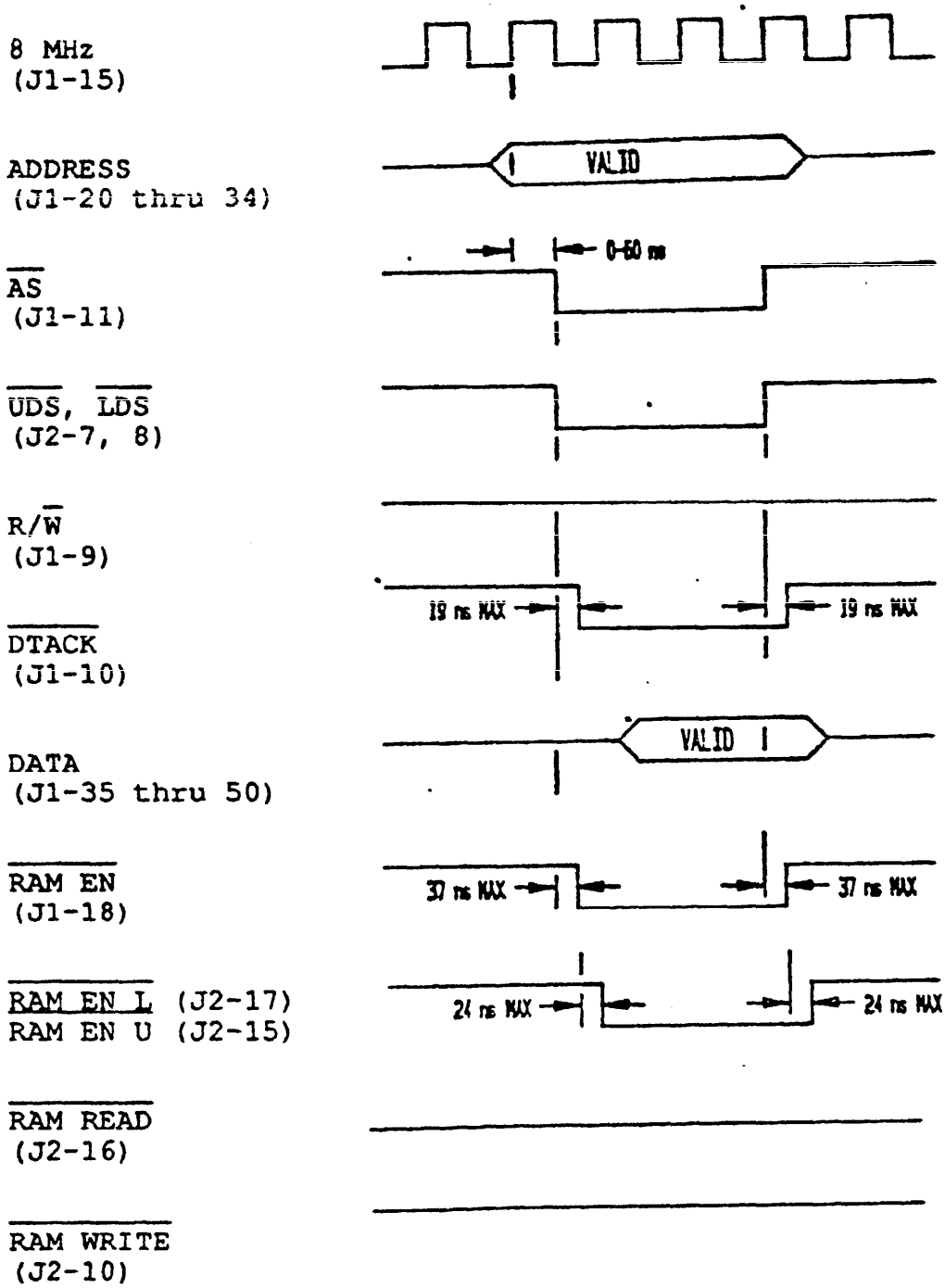


FIGURE 4. RAM read cycle.

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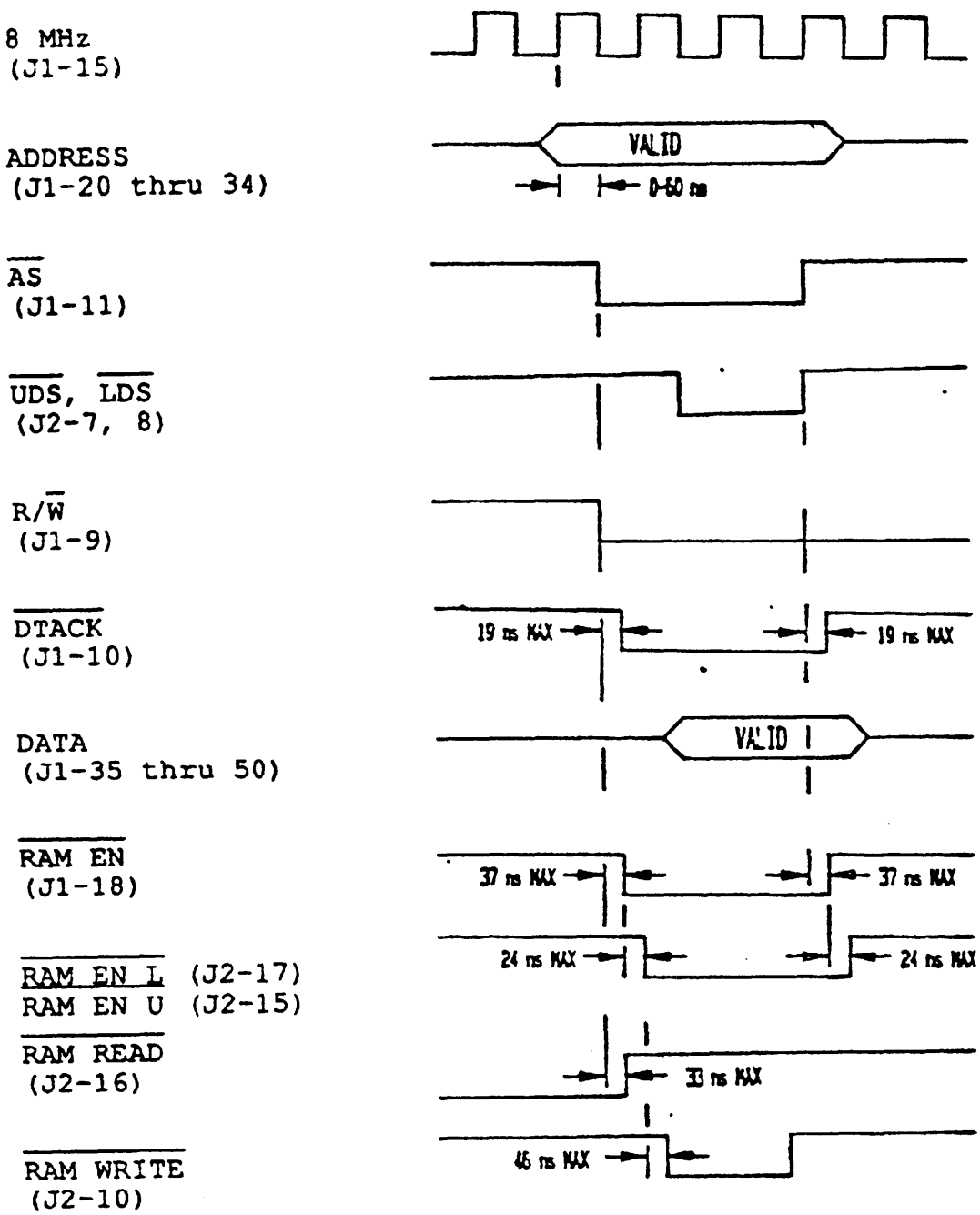


FIGURE 5. RAM write cycle.

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TABLE VIII . power suPPLY and around pins.

PIN No.	SIGnal name
P1-9	+5 V RTN
P1-18	+5 V RTN
P1-23	SCB BOARD IN (+5 V RTN)
P1-32	+5 v
P1-33	+5 v
P1-34	+5 v
P1-35	+5 V RTN
P1-36	+5 V RTN
P1-37	+5 V RTN
P1-38	+
15 P1-39	-15 v
P1-40	+15 v
P1-49	+5 V RTN
P1-58	+5 V RTN
P1-72	+5 v
P1-73	+5 v
P1-74	+5 v
P1-75	+5 V RTN
P1-76	+5 V RTN
P1-77	+5VRTN
P1-78	i 15 V RTN
P1-79	-15 *v*
P1-80	+15 v
P2-18	+5 V RTN
P2-33	+5 v
P2-68	+5 v

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3.6.3 Connector PI outputs. The SCB connector PI outputs shall be TTL per 3.5.3 with output pins as specified in Table X.

3.6.4 Connector PI input/outputs. The SCB connector P1 input/outputs shall be l'"1'L per 3.5.3 with input/output pins as specified in Table XI.

3.6.5 Connector P2 inputs. The SCB connector P2 inputs shall be as listed in Table XII.

3.6.6 Connector P2 outputs. The SCB connector P2 outputs shall be listed in Table XIII.

3.6.7 Connector P2 test points. The SCB connector P2 test points shall be TTL per 3.5.3 with test point pins as specified in Table XIV.

3.6.8 Connector J1 input and outputs. The SCB connector J1 input and output pins shall be TTL per 3.5.3 with input and output pins as specified in Table XV.

3.6.9 Connector J1 test points. The SCB connector J1 test points shall be TTL per 3.5.3 with test points as listed in Table XVI .

3.6.10 **Comector J2 test points. The SCB connector J2 test** points shall **be TTL per 3.5.3 with test points as listed in Table** XVII .

3.6.11 Spare pins. The SCB spare pins shall be as listed in Table XVIII.

3.7 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see Ass).

3.8 Soldering. All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877.

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TABLE IX. Connector **PI** input **pins**.

PIN NO.	SIGNAL NAME
19	B1/2 A6
20	B1/2 A4
21	B1/2 A2
22	SCB ENABLE
24	READ
59	B1/2 AS
60	B1/2 A3
61	B1/2 A1
64	2 DELAY WRITE
69	SYN DTACK

TABLE X. Connector PI output pins.

PIN No.	signal name
25	B3 A4
26	B3 A3
28	CLOCK (800 KHz)
29	syn en
30	r/w
63	CPU INTERRUPT
65	B3 A2
66	B3 Al
70	lds

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Connector PI input /output pins .

1	B1/2 DIS
2	B1/2 D13
3	B1/2 Dll
4	B1/2 D9
5	B1/2 D7
6	B1/2 D5
7	B1/2 D3
8	B1/2 D1
10	B3 D15
11	B3 D13
12	B3 Dll
13	B3 D9
14	B3 D7
15	B3 D5
16	B3 D3
17	B3 D1
41	01/2 D14
42	B1/2 D12
43	B1/2 D10
44	B1/2 D8
45	B1/2 D6
46	B1/2 D4
47	B1/2 D2
48	B1/2 D0
50	B3 D14
51	B3 D12
52	B3 D10
53	B3 D8
54	B3 D6
55	B3 D4
56	B3 D2
57	b3 D0

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TABLE XII. Connector P2 input pins.

PIN NO.	SIGNAL NAME	SIGNAL
3	HAND CONTROL EL HI	± 15 Vdc
4	HAND CONTROL EL LO	± 15 Vdc
6	GUN TACH EL HI	± 15 Vdc
7	GUN TACH EL LO	± 15 Vdc
8	EL TACH OUT	± 15 Vdc
9	CURRENT EL HI	± 15 Vdc
10	CURRENT EL LO	± 15 Vdc
12	MANUAL/EXTERNAL RANGE	TTL per 3.5.3
13	$\overline{\text{U2 PGM}}$	TTL per 3.5.3
14	$\overline{\text{U3 PGM}}$	TTL per 3.5.3
21	+24 V (BTRY) RTN	GROUND
38	HAND CONTROL AZ HI	± 15 Vdc
39	HAND CONTROL AZ LO	± 15 Vdc
41	GUN TACH AZ HI	± 15 Vdc
42	GUN TACH AZ LO	± 15 Vdc
43	AZ TACH OUT	± 15 Vdc
44	CURRENT AZ 1 HI	± 15 Vdc
45	CURRENT AZ 1 LO	± 15 Vdc
46	CURRENT AZ 2 HI	± 15 Vdc
47	CURRENT AZ 2 LO	± 15 Vdc
49	GROUND MODE	0 to +24 Vdc
50	ACTION SWITCH	0 to +24 Vdc
51	FAULT BYPASS	0 to +24 Vdc
52	SP/$\overline{\text{TWD}}$	TTL per 3.5.3
53	S/D 1-BIT	TTL per 3.5.3
54	S/D 2-BIT	TTL per 3.5.3
55	S/D 3-BIT	TTL per 3.5.3
56	S/D 4-BIT	TTL per 3.5.3

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TABLE XIII. Connector P2 output pinsl

pin No.	signal name	SXGNAL
31	GUN COMMAND EL HI	± 10 Vdc
32	GUN COMMAND EL LO	GND
34	SIGHT COMMAND EL HI	± 10 Vdc
35	SIGHT COMMAND EL LO	GND
66	GUN COMMAND AZ HI	± 10 Vdc
67	GUN COMMAND AZ LO	GND
69	SIGHT cOMMAND AZ HI	i 10 Vdc
70	SIGHT COMMAND AZ LO	GND

TABLE XIV. Connector P2 test Point Pins.

PIN NO.	SIGNAL NAME
22	MUX INPUT 10
23	MUX INPUT 11
24	MUX INPUT 12
25	MUX INPUT 13
26	MUX INPUT 14
27	MUX INPUT 15
57	MUX INPUT 1
5 8	MUX INPUT 2
59	MUX INPUT 3
60	MUX INPUT 4
61	MUX INPUT 5
62	MUX INPUT 6
63	MUX INPUT 7
64	MUX INPUT 8
65	MUX INPUT 9

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TABLE XV. Connector J1 input and output Pins.

PIN NO.	SIgNAL name	DIRBCTIon
1	INTERNAL CLOCK DISABLE	INPUT
2	EXT CLOCK	INPUT
3	BUS REQUEST	INPUT
4	BUS GRAND ACK	INPUT
5	B U S	OUTPUT
6	RESET	INPUT

TABLE XVI. Connector Jl teat point pins.

PIN NO.	SIGNAL NAME	PIN No.	SIGNAL NAME	PIN NO.	SIGNAL NAMB
8	128 HZ	20	Al 5	35	D15
9	R/W	21	A14	36	D14
10	DTACK	22	A13	37	D13
11	AS	23	A12	38	D12
12	Ull RESET	24	Al 1	39	Dll
13	3.2 KHz	25	AIO	40	D10
14	800 KHz	26	A9	41	D9
15	8 MHz	27	A8	42	Da
16	16 MHZ	28	A-1	43	D7
17	16 MHZ	29	A6	44	D6
18	RAM EN	30	A5	45	D5
19	ROM EN	31	A4	46	D4
		32	A3	47	D3
		33	A2	48	D2
		34	Al	49	D1
				50	DO

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TABLE XVXI. Connector J2 test point pin.

PIN NO.	SIGNAL NAME	PIN NO.	SIGNAL NAME
1	PR HALT	26	As
2	RESET	27	DEC EN
3	BERR	28	ROM EN
4	HALT RESET	29	RAM EN
5	U10-SET	30	U46 CLR
6	HALT	31	D/A 1 LD
7	UDS	32	D/A 2 LD
8	LDS	33	D/A 3 LD
9	U28-SET	34	D/A 4 LD
10	RAM WRITE	35	D/A DTACK
11	B 1/2 EN	36	ACTION SWITCH
12	DTACK	37	FAULT BYPASS
13	ROM DTACK	38	GND MODE
14	RAM WRITE EN	39	OD
15	RAM EN U	40	[illegible]
16	RAM READ	41	QB
17	RAM EN	42	QA
18	Q	43	A/D IN
19	D/A EN	44	DP
20	U7-SET	45	A/D STR
21	SOC	46	A/D EN
22	EOC	47	SYNCHRO EN
23	A/S STR	48	D/A EN
24	PR HALT	49	HALT EN
25	D/W EN		

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TABLE XVIII. Spare pins.

CONN. PIN No.
P1-27
P1-31
P1-62
P1-67
P1-68
P1-71
P2-1
P2-2
P2-5
P2-11
P2-15
P2-16
P2-17
P2-19
P2-20
P2-28
P2-29
P2-30
P2-36
P2-48
J1-7
J2-50

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or **purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the** contractor may use his own or any other facilities suitable for the performance **of** the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all retirements of Sections 3 and 5. The inspection set forth in this specification shall **become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted** to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4,3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspection (see 4.5).

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4.2 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50°F to +90"F
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches Hg

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantitv
SCB Assembly	9360875	3

4.3.2 Inspections to be Performed. As determined by the Government, the first article assemblies components, and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Qua1ity conformance inspection.

4.4.1 Inspection lot formation. The term 'inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. units of product selected for inspection shall represent only the inspection lot from whit'n they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one

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and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance provisions (QAP) listed in the Technical **Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.**

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, **as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3).** When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

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4.4.2.3 Functional tests. The tests in Table XIX shall be performed on a 100% basis.

TABLE XIX. Functional tests {100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical	None	N/A
Major		
101. Electrical	3.4	4.7.2
102. Performance	3*5	4.7.3
Minor	None	NIA

4.5 Special samping. One SCB out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table XIX and shall then meet the requirements and tests in Table xx. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XX. Special Sampling.

CHARACTERISTIC	-RequirEmENT	TEST PROCEDURE
Critical	None	N/A
Major		
103. vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3*3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3*3.5	4.7.1.5
Minor	None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test, as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection Equipment.

4.6.1 Government-furnished inspection equipment. Where the contract **provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.**

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design The contractor shall design and SUpply inspection equipment compatible with the 'Methods of Inspection. specified in 4.7 of this specification and with the component inspection procedures specified in 'Examination" and 'Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

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4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table XXI (Test equipment list) shall be utilized in the performance of the applicable test methods of inspection as specified in 4.7. The following items, or equivalent, are required to test the SCB and should be assembled at the M/USM-465A Test Station prior to test execution. Proper operation of interface devices ad test equipment shall be verified before using same to run diagnostic tests.

TABLE XXX. Test equipment list

ITEM NUMBER	DESCRIPTION
1. Interface Device TPS Adapter Board B	9397371
2. Cable Assembly WI (Qty 2)	9397497
3. Test Plug P6	9397523
4. Test Plug P7	9397524
5. Test Plug P8	9397525
6. Test Plug P9	9397526
7. Digital Card Tester	AN/USM-465A
8. Comcode Program Tape	Version 2.0
9. Test Program Set Manual	TPS 9361337
10. Test Program Set Tape	TPT 9361337

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. This test shall be performed as specified in Method 514, Category 8, Procedure I of MIL-STD-810 and paragraph 3.3.1 herein. In step 1 of Procedure 1, only steps 4 through 8 of paragraph 11-2.1 apply. The SCB shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment- After vibration, the SCB shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section 11 of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms +* 10% duration.

c. The maximum g shall occur at 5.5 milliseconds +| lo%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the SCB shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude operating. This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

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4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual extinction for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 oscillator and counting circuit. The oscillator and counting circuit shall perfom as per 3.5.4 and shall be tested in accordance with oscillator and Counter Circuit, test 0003, of the SCB Test Program Set (TPS), drawing 9361337.

4.7.3.2 1/0 address decoder. The 1/0 address decoder circuit shall perfom as per 3.5.6 and shall be tested in accordance with 1/0 Address Decoder and DTACK Logic, test 0005, of TPS 9361337.

4.7.3.3 AID converter interface. The AID converter interface circuit shall perform as per 3.5.7 and shall be tested in accordance with Analog Interface Test and A/D-D/A, or Control Logic and Dynamic RAM/ROM Test, tests 0012 and 0013, 9361337-

4.7.3.4 Halt control circuit. The halt control circuit shall perform as per 3.5.10 and shall be tested in accordance with Halt Control, Bus Error Logic and CPU Interface Logic, tests 0006 and 0007, of TPS 9361337.

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4.7.3.5 Control panel status. The control panel status circuit shall perform as per 3.5.8 and shall be tested in accordance with Mode Control Interface Logic, test 0009, of TPS 9361337.

4.7.3.6 D/A converters circuit. The D/A converters circuit shall perform as per 3.5.9 and shall be tested in accordance with D/A Control Logic and DC Output, test 0011, of TPS 9361337.

4.7.3.7 Power requirements. The power requirments shall be as specified in Table I and shall be tested in accordance with SCB Identification and Power Supplies, tests 0001 and 0002, Of TPS 9361337.

4.7.3.8 Input/output requirements" The input/outPut requirements shall meet the requirements of 3.6 and shall be tested in accordance with Synchro Board Interface, test 0008, of TPS 9361337.

4.7.3.9 Reset circuit. The reset circuit shall perfom as per 3.5.3 and shall be tested in accordance with PUll-up Resistor and Reset Circuit, test 0004, of TPS 9361337.

4.7.3.10 Mernory requirements. The memory requirements shall be as specified in 3.5.11 and shall be tested in accordance with Static RAM/ROM Access, test 0010, of TPS 9361337.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.8.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360875. The level of protection shall be as specified in the procurment document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The SCB circuit card assembly shall be used in the fire control processor in the VADS M163A2 self-propelled and M167N towed vehicles.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requireent if other than specified in Section 5.

f. Serialization requirements, if applicable.

g" Certificate of conformance for each lot or shipment of product.

6.3 Submission by the contractor.

a. Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

b. Alternative quality conformance provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S/ Army Armament Research, Development and Engineering Center (ARDEC) may also Include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.5 Subject term (key word) listing.

Fire **Control Processor (FCP)**
Vulcan Air Defense Systm (vADS)

6.6 **Changes** from Previous issue. The previous issues were for ARDEC internal use only. TheY were not forwarded to Naval Publications and Forms Center **(NPFC) for publication.** **This issue** is initial public release for unlimited distribution.

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70570C	12 April 1993

3. DOCUMENT TITLE: Circuit Board, Servo Controller: 9360875

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Barbara Irwin	(1) Commercial: (201) 724-6674	(2) AUTOVON: 880-6674
c. ADDRESS (Include Zip Code) US Army ARDEC (SMCAR-RAC-S) Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office [illegible] Telephone (703) 756-2340 AUTOVON 289-2340	

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